

VALOGNES, France

WMO: 070390

Lat: 49.517N

Lon: 1.500W

Elev: 61

StdP: 100.59

Time Zone: 1.00 (EUC)

Period: 99-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	-1.1	-4.9	2.5	0.0	-3.5	2.8	0.8	14.3	9.2	12.4	8.6	1.3	10	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	25.5	19.5	23.6	18.6	22.0	17.4	20.0	24.6	18.9	22.6	18.0	21.1	2.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.3	22.2	17.4	12.6	20.8	16.7	12.0	19.5	57.6	24.7	53.9	22.8	51.0	21.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.1	8.2	7.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-5.2	29.1	1.6	2.8	-6.4	31.1	-7.3	32.7	-8.2	34.3	-9.4	36.3
			-5.5	22.2	1.6	1.6	-6.6	23.3	-7.6	24.2	-8.4	25.1	-9.6	26.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	6.1	6.1	7.3	9.5	12.5	15.1	16.6	16.8	15.0	12.0	8.6	6.2
	DBStd	4.83	3.08	2.95	2.94	2.75	2.54	2.34	2.17	2.14	2.52	2.85	2.71	3.22
	HDD10.0	568	123	112	92	41	6	0	0	0	0	14	57	123
	HDD18.3	2699	378	342	342	264	182	102	64	57	104	196	291	376
	CDD10.0	937	3	3	8	26	83	152	203	211	151	76	17	6
	CDD18.3	27	0	0	0	0	1	4	9	9	4	0	0	0
	CDH23.3	188	0	0	0	0	8	31	64	62	22	1	0	0
	CDH26.7	33	0	0	0	0	1	4	8	17	2	0	0	0
Wind	WSAvg	3.2	3.8	3.9	4.0	3.7	3.4	2.9	2.8	2.6	2.5	2.7	2.9	3.6
Precipitation	PrecAvg	901	101	76	61	60	52	49	49	59	61	101	114	119
	PrecMax	1159	182	162	157	156	115	111	103	151	121	216	286	254
	PrecMin	641	18	12	12	7	10	12	15	10	9	27	32	14
	PrecStd	129	47	43	32	38	27	28	24	35	28	48	57	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.7	16.4	20.4	24.3	26.5	28.1	29.0	26.1	21.2	16.0	13.7
		MCWB	11.5	10.6	11.9	14.5	18.2	19.6	19.6	23.3	19.2	16.9	14.6	12.4
	2%	DB	12.1	12.2	14.5	17.6	21.1	23.7	25.5	24.6	23.1	18.4	14.5	12.7
		MCWB	11.0	10.4	11.3	12.8	16.1	18.5	19.2	19.6	18.3	15.6	12.9	11.7
	5%	DB	11.2	11.4	12.9	15.6	19.1	21.6	23.3	22.7	21.2	17.1	13.7	12.0
		MCWB	10.1	9.9	10.7	11.9	15.1	16.8	18.0	18.4	17.3	15.0	12.1	10.9
	10%	DB	10.5	10.6	11.8	14.1	17.1	20.0	21.5	21.1	19.4	16.2	12.9	11.2
		MCWB	9.4	9.3	10.1	11.2	13.8	15.9	17.0	17.1	16.3	14.5	11.5	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	11.7	12.9	15.2	18.8	20.5	21.0	23.3	20.2	17.7	15.3	12.8
		MCDB	12.6	12.8	15.4	18.6	23.8	25.1	26.4	28.7	24.3	19.7	16.0	13.2
	2%	WB	11.2	11.0	11.8	13.5	16.6	18.8	19.6	20.1	18.7	16.2	13.4	11.9
		MCDB	11.9	11.8	13.4	16.7	20.3	23.0	24.5	24.1	21.9	17.6	14.1	12.5
	5%	WB	10.4	10.3	11.1	12.3	15.3	17.4	18.5	18.7	17.8	15.5	12.5	11.1
		MCDB	11.1	11.2	12.5	14.9	18.5	20.7	22.5	21.8	20.3	16.7	13.4	11.9
	10%	WB	9.6	9.5	10.3	11.5	14.2	16.4	17.5	17.8	16.9	14.8	11.6	10.1
		MCDB	10.4	10.5	11.5	13.7	16.7	19.3	20.6	20.4	18.8	16.0	12.7	11.1
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.6	6.4	7.4	7.5	8.4	7.9	8.1	8.4	7.0	6.3	5.2
		MCDBR	4.4	5.9	8.1	10.0	11.0	11.3	11.8	10.9	10.9	7.9	6.3	4.6
		MCWBR	4.2	4.7	5.6	6.1	6.9	6.8	6.6	6.3	6.4	5.7	5.0	4.7
	5% WB	MCDBR	4.0	4.9	6.9	9.0	10.1	10.1	10.3	10.1	9.5	6.9	5.7	4.5
		MCWBR	4.0	4.5	5.2	5.7	6.7	6.6	6.0	6.2	5.9	5.3	5.1	4.7
Clear-Sky Solar Irradiance	taub	0.332	0.348	0.381	0.393	0.399	0.395	0.389	0.383	0.372	0.361	0.336	0.326	
	taud	2.412	2.395	2.309	2.296	2.329	2.371	2.388	2.426	2.438	2.454	2.443	2.380	
	Ebn at Noon	711	791	821	851	860	866	865	854	825	767	701	657	
	Edh at Noon	65	83	107	120	121	117	113	104	93	78	63	59	
All-Sky Solar Radiation	RadAvg	0.88	1.61	2.81	4.44	5.25	5.84	5.63	4.66	3.57	2.00	1.07	0.72	
	RadStd	0.08	0.19	0.30	0.38	0.40	0.56	0.36	0.31	0.22	0.17	0.13	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% DP	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page